

# COMP3004: Designing Intelligent Agents

Topic Approval Submission

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## Project Description

This project designs and evaluates autonomous liability trading agents within the Rotman Interactive Trader (RIT) LT3 simulation, where agents must assess incoming institutional block trade tenders, make accept/decline decisions, and profitably unwind accumulated positions within a five-minute trading session subject to net and gross position limits.

The central research question is: *Can a genetically parameterised agent architecture, where strategy parameters are evolved offline against a P&L-based fitness function, outperform hand-coded reactive and deliberative baseline agents in live RIT LT3 sessions?*

Three agents are implemented: A **reactive agent** applying fixed spread thresholds with immediate market order unwinding; a **deliberative agent** reasoning over the full visible order book (up to 20 levels via the RIT API), estimating slippage, and executing a planned mixed limit/market unwind schedule; and a **GA-evolved agent** sharing the deliberative architecture but with key parameters (including spread acceptance threshold, liquidity depth ratio, market order aggressiveness, and time decay urgency) optimised via a genetic algorithm evaluated against synthetic order books parameterised from observed RIT session data.

All three agents are compared across repeated live sessions on P&L, end-of-session residual position, and tender acceptance quality using non-parametric statistical tests.